Overview

This project highlights a real-world ETL process built in Excel to clean and transform a CRM data export for use in an automated email campaign. The goal was to drive End of Course Survey (EOCS) participation by targeting currently active students with tailored communication.

Challenges

The CRM export contained student name, number, and email on one worksheet—ideal for automation.

However, the data appeared encrypted or obfuscated, likely to minimize exposure risk in the event of a data breach.

This introduced hidden characters and inconsistent formatting that broke Excel features like filters and VLOOKUP.

I also needed to integrate notes from previous manually maintained reports for context.

ETL Process

Extract:

Received raw CRM export from a data analyst.

Transform:

Copied the data into a clean worksheet to strip formatting.

Applied CLEAN() and TRIM() to remove hidden characters and excess whitespace.

Used VLOOKUP to import relevant notes from historical tracking.

Filtered for currently active students who had not received EOCS-specific outreach in the past year.

Load:

Built an HTML email template directly within Excel.

Executed an automated email campaign to over 200 filtered student records using Outlook scripting.

Outcome

Successfully transformed unusable CRM data into a clean, actionable dataset. Enabled strategic outreach that had previously been impossible, improving EOCS participation and student engagement. This project demonstrates hands-on ETL experience, Excel data wrangling, logic-based filtering, and scalable automation.

VBA Email Automation Script

```
Sub SendEmails()

    Dim OutlookApp As Object

    Dim OutlookMail As Object

    Dim ws As Worksheet

    Dim i As Long

    Dim emailAddress As String

    Dim subjectLine As String

    Dim emailBody As String



    ' Use the active worksheet instead of hardcoding the name

    Set ws = ActiveSheet
```

```
Set OutlookApp = CreateObject("Outlook.Application")



For i = 2 To ws.Cells(ws.Rows.Count, 1).End(xlUp).Row ' Assuming row 1 has headers

    emailAddress = ws.Cells(i, 1).Value

    subjectLine = ws.Cells(i, 2).Value

    emailBody = ws.Cells(i, 3).Value



    If emailAddress <> "" Then

        Set OutlookMail = OutlookApp.CreateItem(0)

        With OutlookMail

            .To = emailAddress

            .Subject = subjectLine
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            .HTMLBody = emailBody

            .Send

        End With

        Application.Wait Now + TimeValue("00:01:00") ' Wait 1 minute before next email

    End If

  Next i



  MsgBox "Emails sent successfully!"

End Sub
```